

DANA 4830 | SUMMER 2026

A1 - Exploratory Data Analysis (EDA) for Feature Selection and Dimensionality Reduction in Healthcare

Assignment Overview

This assignment develops your ability to conduct rigorous Exploratory Data Analysis (EDA) on large-scale, real-world healthcare data as a precursor to feature selection and dimensionality reduction. Healthcare datasets are inherently complex, characterised by high dimensionality, mixed data types, substantial missingness, and class imbalance

Dataset Eligibility Requirements

You must identify and use a publicly accessible healthcare dataset that satisfies ALL of the following mandatory criteria.

#	Criterion	Specification
1	Dataset domain	Healthcare or clinical (e.g. patient records, clinical trials, hospital admissions, disease registries, electronic health records)
2	Minimum row count	More than 500,000 rows (observations / patient records)
3	Minimum feature count	More than 30 features (columns), not including any row identifier
4	Feature type composition	Must contain both categorical features (e.g. diagnosis codes, gender, ethnicity, comorbidity flags) and numerical features (e.g. vitals, lab values, age, dosage)
5	Public accessibility	Freely accessible without a data purchase. Institutional login or registration (e.g. PhysioNet credentialing) is permitted provided the dataset is openly available to the public.

Note that: MIMIC IV is a large dataset but when you extract data to your relevant matter, the size of the data becomes much smaller. So the above criteria is applied for the case when you use the dataset to conduct EDA and perform machine learning models.

Possible examples are below.

<https://www.kaggle.com/datasets/imtkaggleteam/synthetic-medical-dataset>

<https://www.kaggle.com/datasets/maheshdadhich/us-healthcare-data>

Mark allocation

The assignment is divided into four graded sections totalling 100 marks. All analysis must be performed in Python within a **Jupyter Notebook and must be fully reproducible**. Accompany your code with clear narrative explanations of every analytical decision and finding. The mark distributions are below.

Section A: Dataset Identification & Justification [15 Marks]

Section B: Data Dictionary [15 Marks]

Section C: Data Quality Assessment (including out-of-range values, inconsistent coding, and duplicate records, hypothesise whether missingness is likely MCAR, MAR, or MNAR) [15 Marks]

Section D: Exploratory Data Analysis [50 Marks]

D1 - Univariate Analysis: Numerical Features [12 Marks]

D2 - Univariate Analysis: Categorical Features [10 Marks]

D3 - Bivariate Analysis: Numerical-Numerical Relationships [14 Marks]

D4 - Bivariate Analysis: Feature-Target Relationships [14 Marks]

Section E: Problem Formulation [5 Marks]. The sample of how to state the problem is herewith:
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9274320/>

Submit the following two files via the BrightSpace

- Jupyter Notebook (.ipynb) — containing all code, visualisations, and narrative explanations. The notebook must execute fully from top to bottom without errors when run on a clean Python environment.
- PDF Report - export of the executed notebook (File > Download as PDF), ensuring all outputs and figures are visible.

Code and Reproducibility Standards

- Set a random seed (e.g. `random_state=42`) for all stochastic operations.
- Use a `requirements.txt` or a `conda environment.yml` to document library versions or instruction to run `uv` command.
- All figures must include axis labels, a title, and a legend where applicable.
- All statistical test results must include the test statistic and p-value.
- Narrative cells must precede each code cell explaining what the code does and why.